

Architecture & Design Document

1. Project Overview

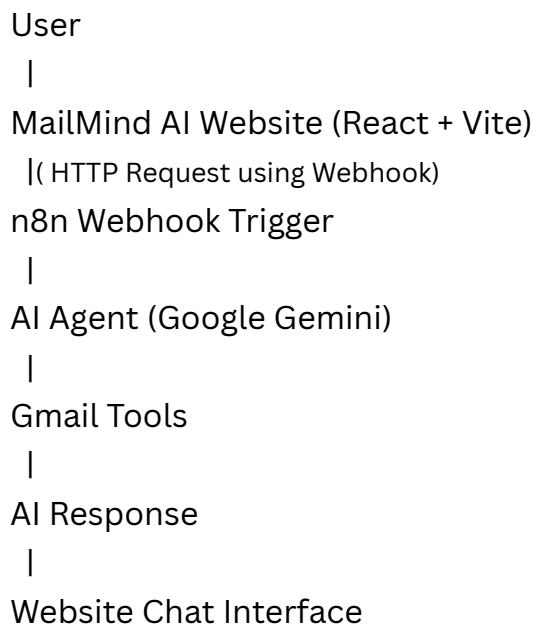
MailMind AI is an AI-powered email assistant that helps users manage their Gmail accounts using natural language commands.

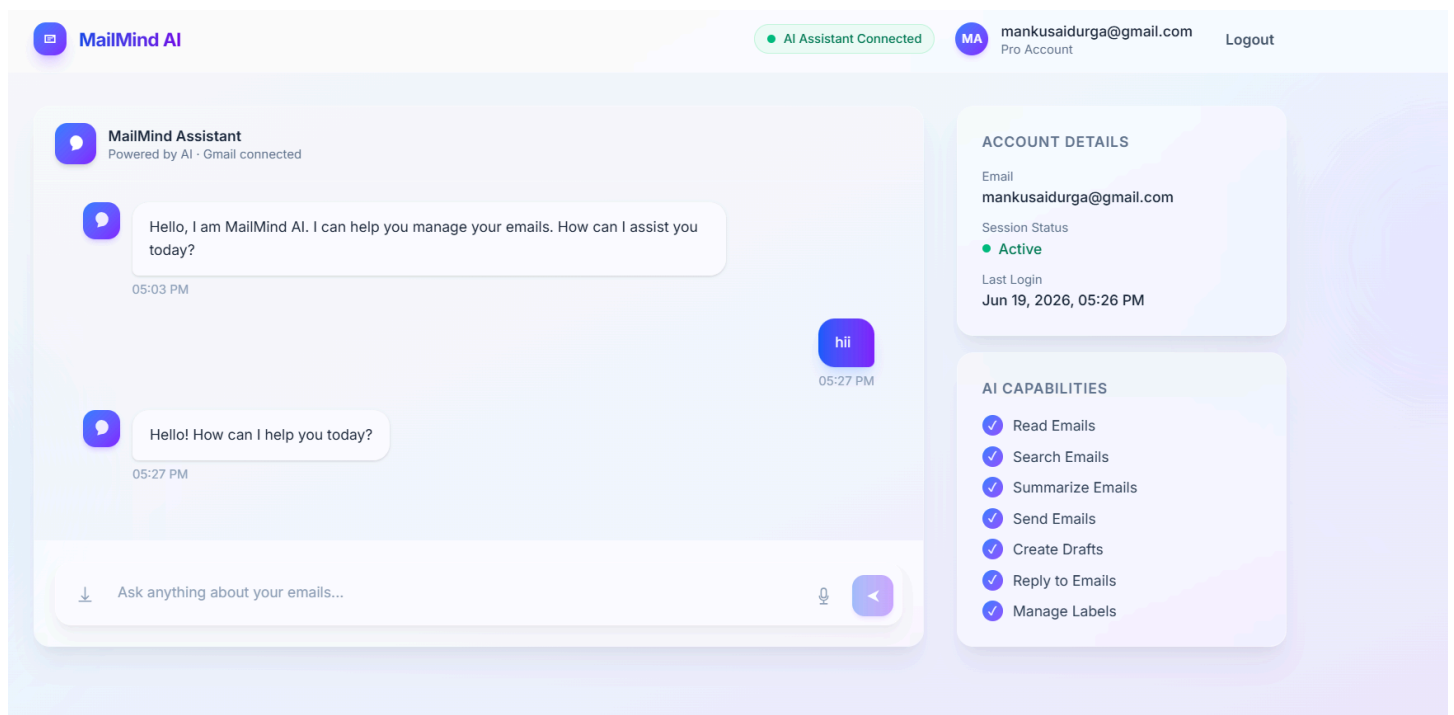
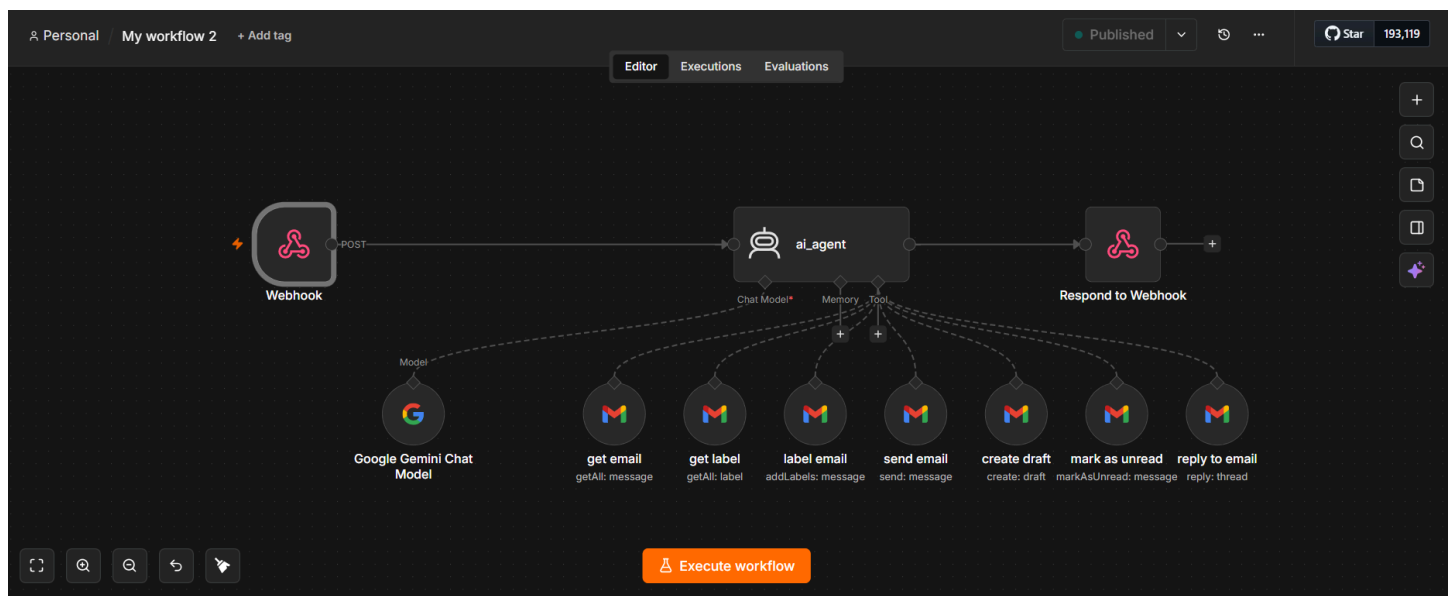
The project was built in two major parts:

- **n8n AI Workflow** – Created an AI agent using Google Gemini, memory, and Gmail tools to perform email operations such as **reading emails, creating drafts, sending emails, adding labels, and managing messages.**
- **Web Application** – A user-friendly chat interface was created using Cursor with React, TypeScript, and Vite. Later, the complete project code was shifted to VS Code for manual development and final integration.

2. System Architecture

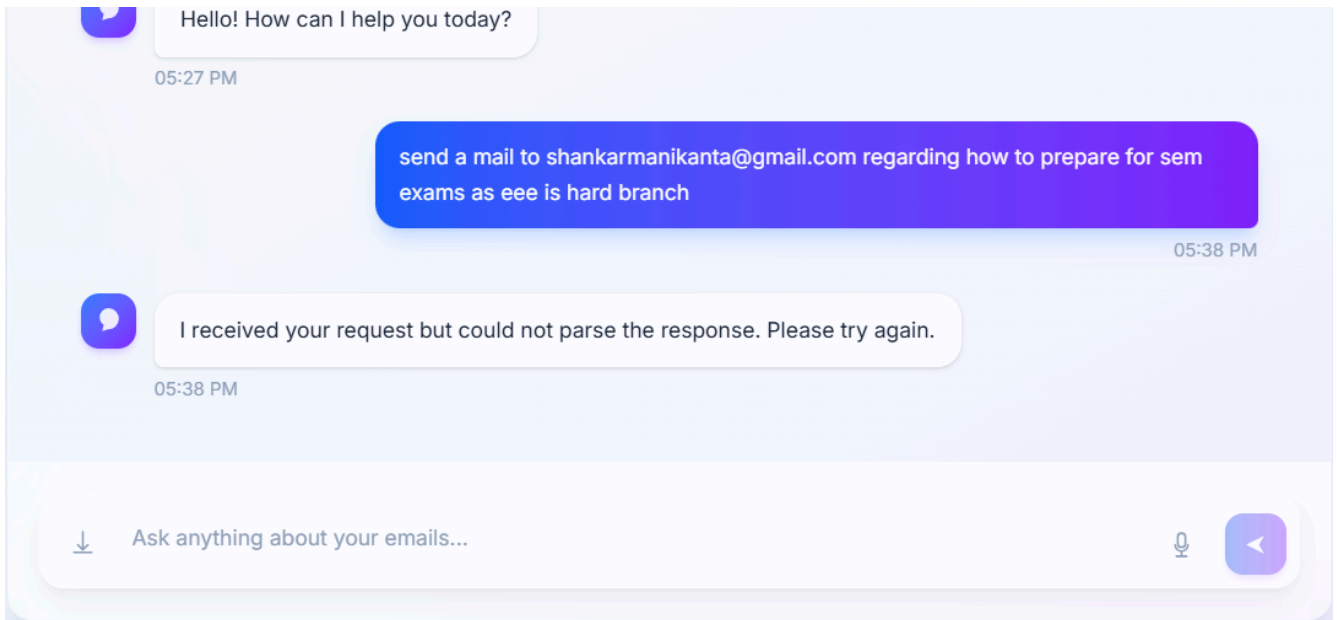
The complete flow of MailMind AI is:



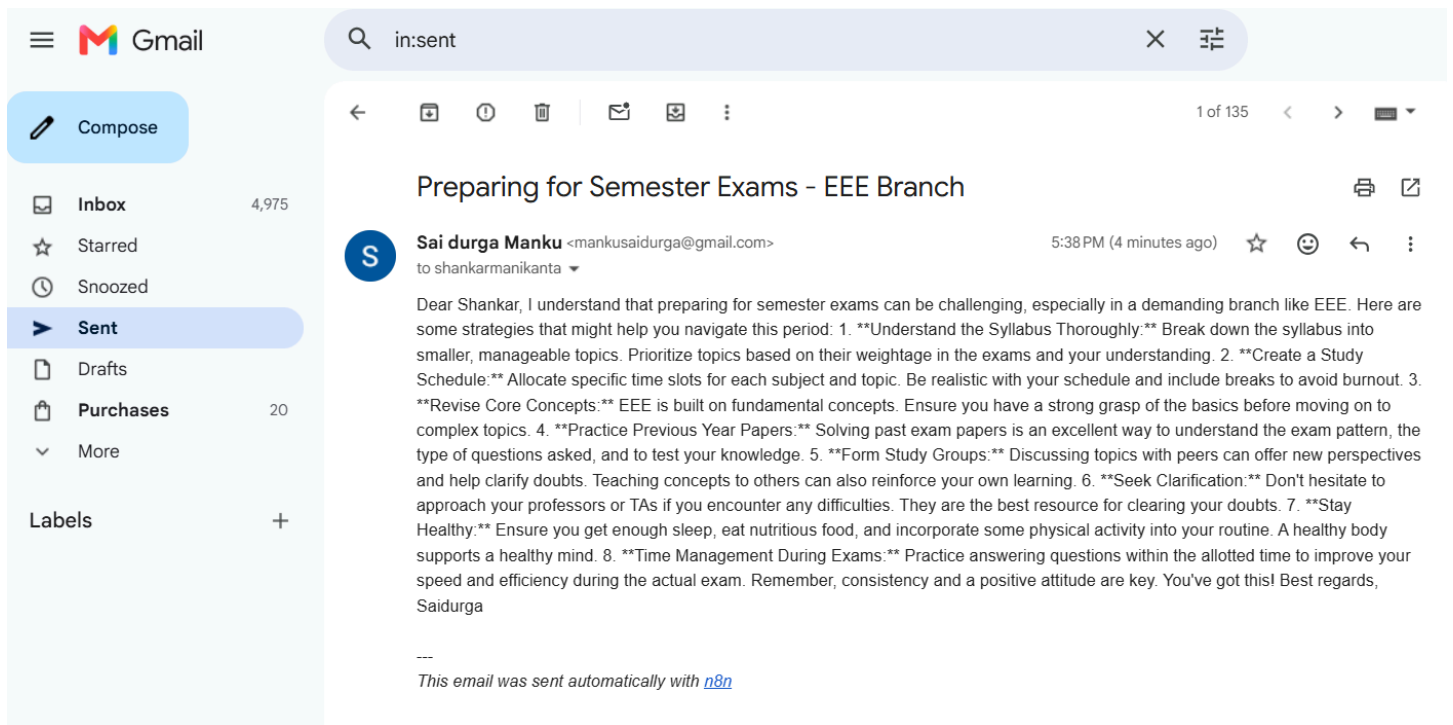


3. WorkFlow

INPUT(given instruction to bot)



OUTPUT(ai bot sent email to shankar)



4. Technologies Used

Frontend

- React
- TypeScript
- Vite
- HTML
- CSS

AI & Backend Automation

- n8n
- Google Gemini AI
- Gmail API
- Webhooks

Development Tools

- Cursor
- VS Code
- GitHub
- Vercel

5. Conclusion

MailMind AI combines AI, automation, and a modern web interface to simplify email management. The project demonstrates the integration of n8n AI workflows with a React-based frontend through Webhooks, creating a complete end-to-end AI email assistant.